INCH-POUND
MIL-DTL-24231/7F(SH)
26 April 2013
SUPERSEDING
MIL-C-24231/7E(SH)
15 May 1987

DETAIL SPECIFICATION SHEET

CONNECTOR, SINGLE CABLE, TYPE III, SEVEN-CONDUCTOR

This specification is approved for use by the Naval Sea Systems Command and is available for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-DTL-24231.

REQUIREMENTS:

1. Dimensions and configuration: see figures 1 through 7.

2. Full or partial threads shall not extend more than 0.083 inch beyond the ¾-inch thread length.

3. The gland nut shall be tightened until the packing tightens against the wires. This can be determined by moving the wires gently back and forth until the wire is seized by the packing. The gland nut shall then be turned approximately one-fourth of a turn more to hold the wires in place. Do not overtighten the gland nut or attempt to tighten down metal to metal, as this can damage the wires.

4. Refer to MIL-DTL-24231/5 for alternate method of manufacturing connector body and installing guide pin M24231/5-071.

5. The retainer plates and the retainer plate assemblies shall be manufactured from nickel-copper alloy in accordance with Class A, Form 1, hot-finished condition of QQ-N-281 with a minimum yield strength of 40,000 lbf/in^2 at 0.2 percent offset, or nickel-copper alloy in accordance with Class A, Form 1, cold-drawn condition of QQ-N-281 with a minimum yield strength of 50,000 lbf/in^2 at 0.2 percent offset.

6. For welding of the pin, see MIL-DTL-24231/5.

7. Retainer plate, packing, and retainer plate assembly for other special conductor arrangements (for example, LS2SWU-1 and B-16 wire combinations) shall meet the following requirements:

   a. Where LS2SWU-1 cable is required, the hole size shall be 17/64 diameter.
   b. Positional tolerancing for all holes shall be ⌖ | Ø.010.
   c. Holes shall not be located closer than 0.010 from the minimum gland nut inside diameter of part number M24231/5-072 and the retainer plate position pin holes of part number M24231/7-030.
   d. Minimum material between holes shall be 0.031.
   e. Protective sleeving shall be sized to suit the number of cables required where use of the specified sleeving, item 18, is inadequate.

8. All dimensions are in inches.

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9. Tolerances shall be as follows, or as stated:

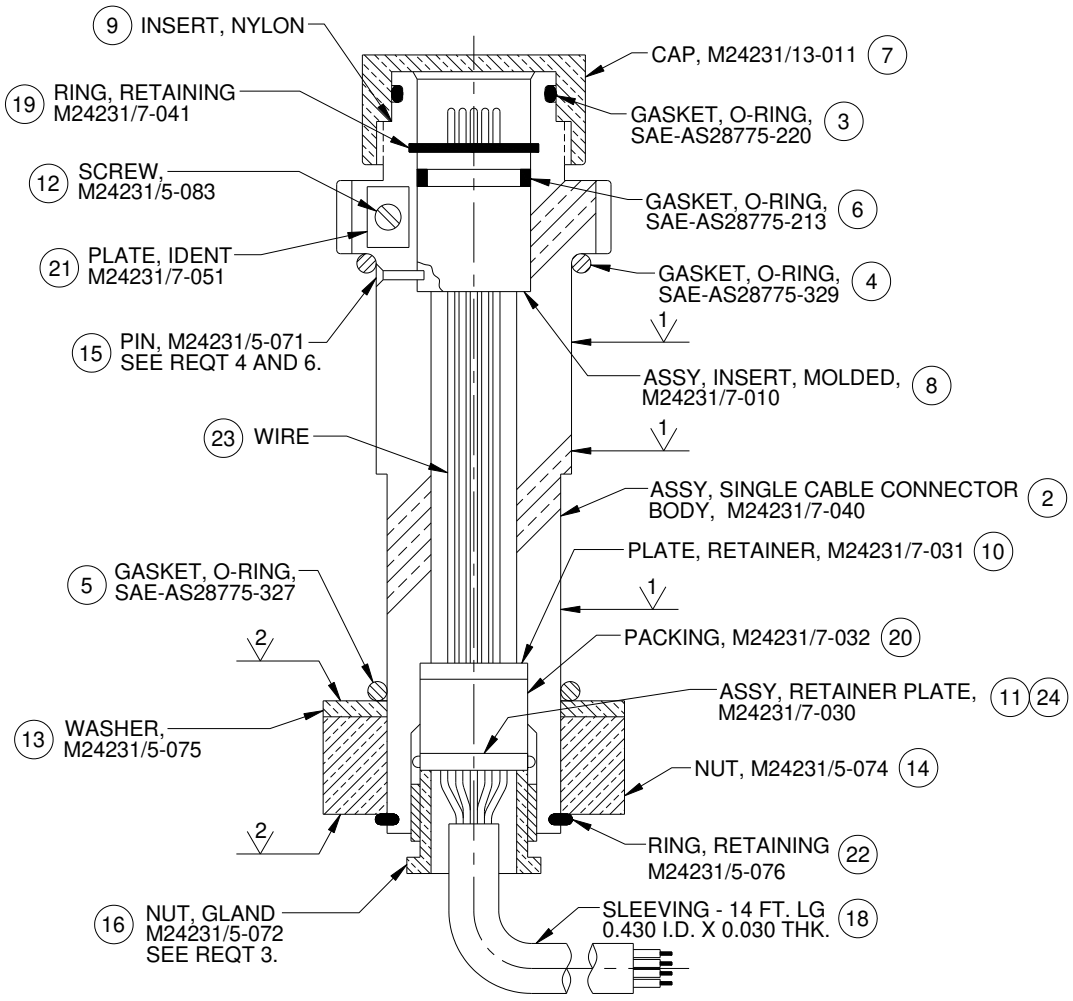
a. Decimals – ±0.005 inch.

b. Fractions – ±1⁄64 inch.

c. Angular dimensions – ±½ degree.

d. Tolerances for rubber packing should follow the guidance provided in table XLI (RMA Drawing Designation A2 Precision) of MIL-HDBK-149.

10. The flange outside diameter is intended to coincide with commonly available 2¾-inch bar stock. The dimension may vary after machine clean-up.



1/ HARDNESS READING TO BE TAKEN AT THESE POINTS.

2/ HARDNESS READING TO BE TAKEN AT THIS FACE AT 120-DEGREE INTERVALS.

FIGURE 1. Connector, seven-conductor – M24231/7-001, symbol no. 517.1. (1)

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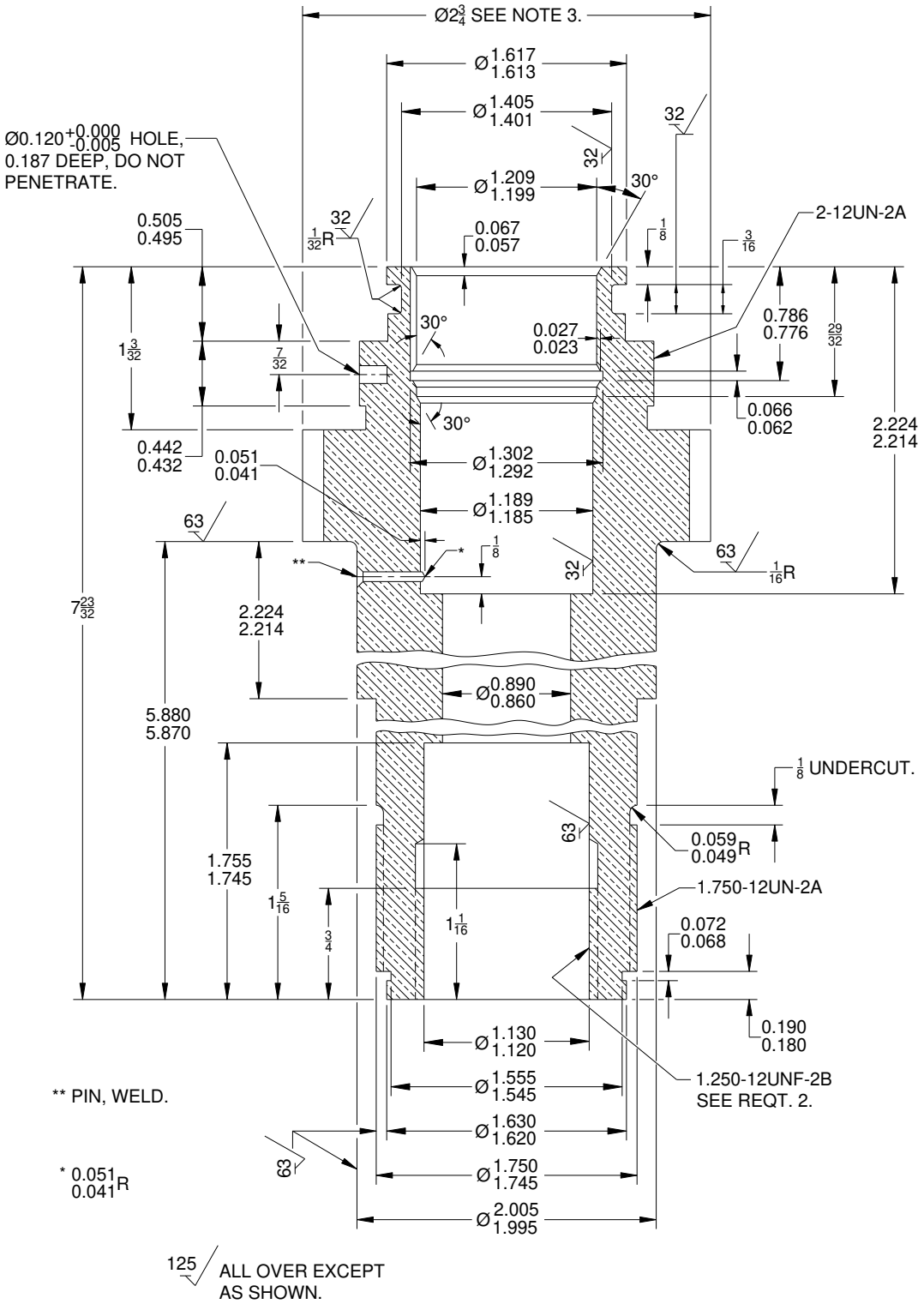


FIGURE 2. Assembly, single cable connector body – M24231/7-040. (2)

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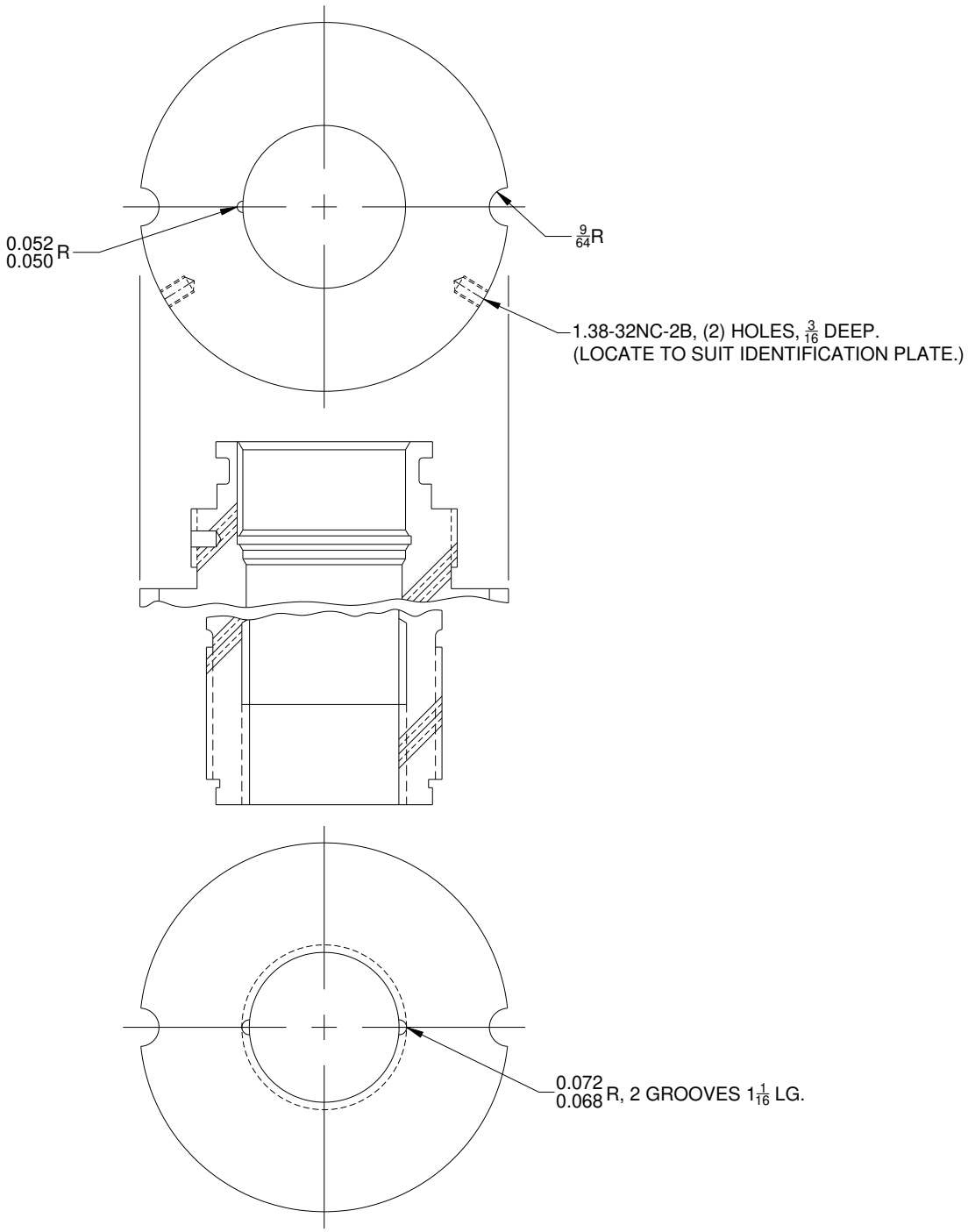


FIGURE 2. Assembly, single cable connector body – M24231/7-040 – Continued. (2)

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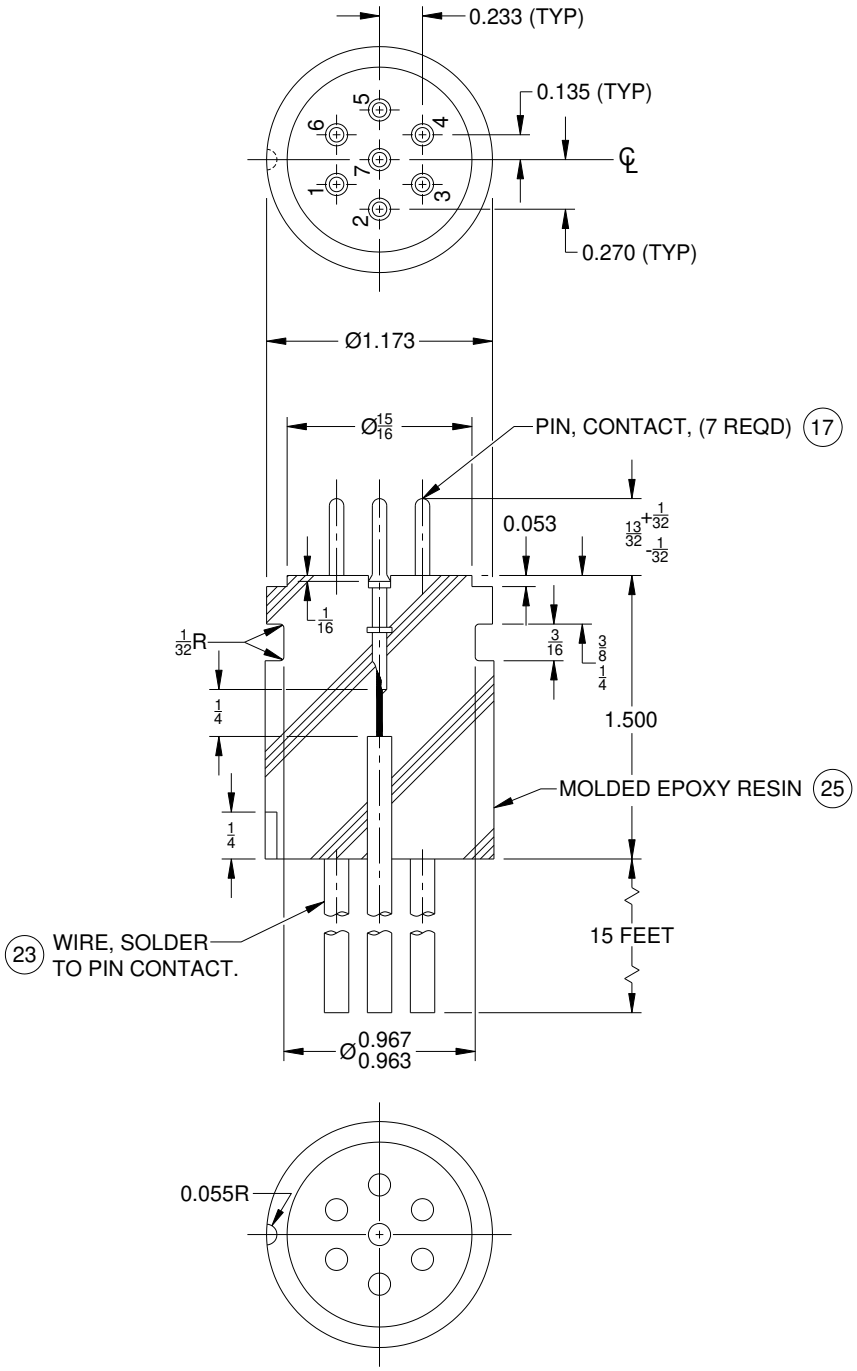


FIGURE 3. Assembly, insert, molded - M24231/7-010. (8)

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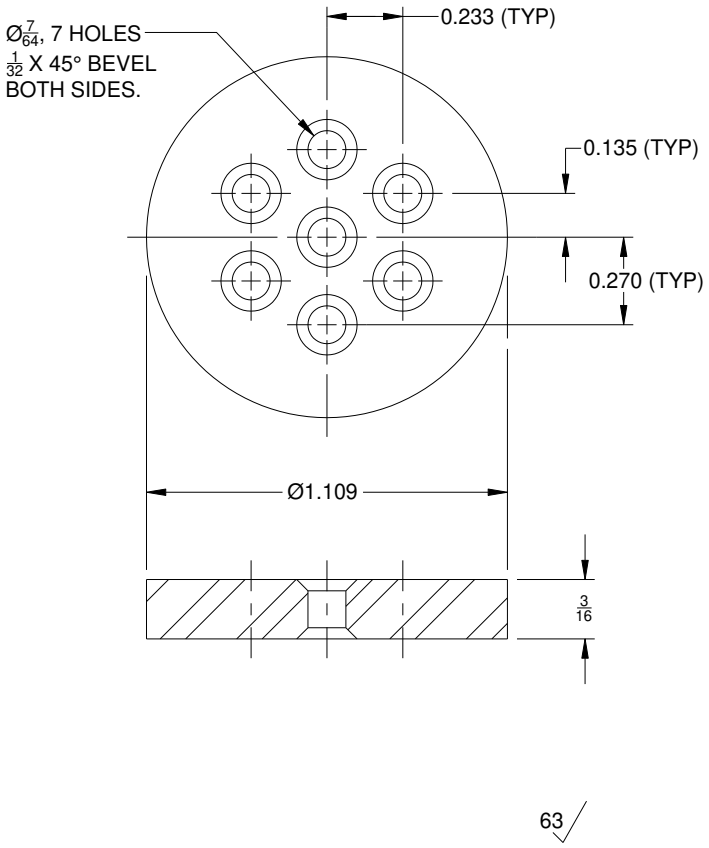


FIGURE 4. Plate, retainer - M24231/7-031. (10)

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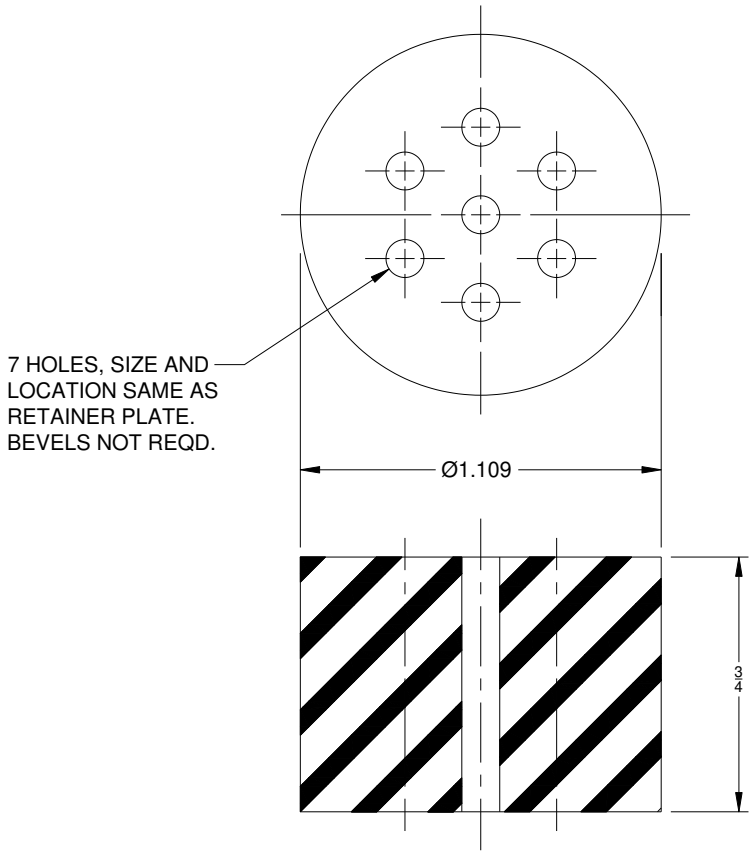


FIGURE 5. Packing - M24231/7-032. (20)

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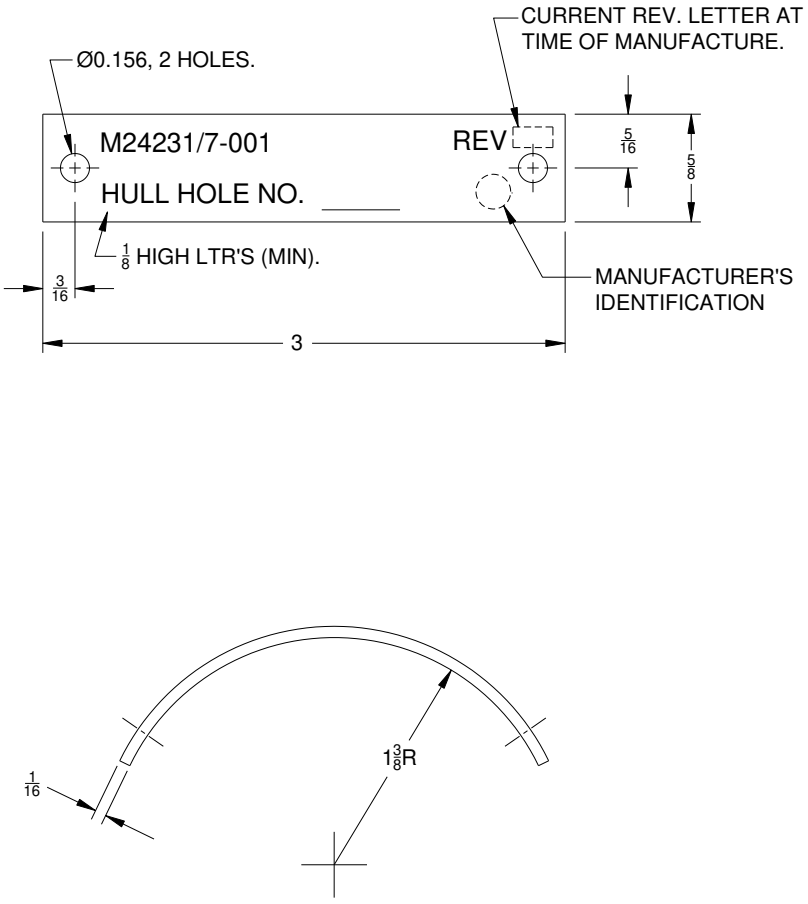


FIGURE 6. Plate, identification – M24231/7-051. (21)

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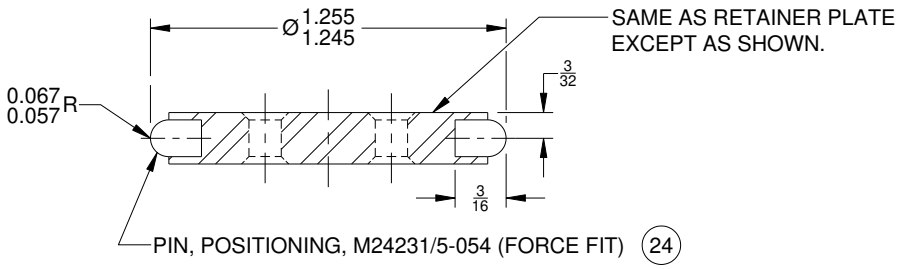


FIGURE 7. Assembly, retainer plate – M24231/7-030. (11)

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TABLE I. Parts list.

Item no. 1/	Description 2/	Quantity for M24231/7- 3/	Material/specification	Remarks
		001		
1	Connector assembly, single cable, symbol no. 517.1	1	-----	See figure 1.
2	Connector, body assembly, single cable	1	Ni-Cu-Al, QQ-N-286	See figure 2 and continuation of figure 2.
3	Gasket, O-ring	1	SAE-AS28775-220 4/	Lubricate with silicone compound, SAE-AS8660.
4	Gasket, O-ring	1	SAE-AS28775-329 4/	See item 3.
5	Gasket, O-ring	1	SAE-AS28775-327 4/	See item 3.
6	Gasket, O-ring	1	SAE-AS28775-213 4/	See item 3.
7	Cap, pressure-proof	1	M24231/13-011	See figure 1.
8	Insert, assembly, molded	1	See items 17, 23, and 25	See figure 3.
9	Insert	1	M24231/5-077	See figure 1.
10	Plate, retainer	1	Ni-Cu, Class A, QQ-N-281	See figure 4 and requirement 5.
11	Assembly, retainer plate	1	Ni-Cu, Class A, QQ-N-281	See figure 7 and requirement 5.
12	Screw	2	M24231/5-083	See figure 1.
13	Washer	1	M24231/5-075	See figure 1.
14	Nut	1	M24231/5-074	See figure 1.
15	Pin, guide	1	M24231/5-071	See figure 1.
16	Nut, gland	1	M24231/5-072	See figure 1.
17	Pin, contact	7	MIL-DTL-24231	See figure 3.
18	Sleeving, 0.430 id, 0.030 thick	14’	Type F, Form U, Category 1, color clear, MIL-I-631	See figure 1.
19	Ring, retaining	1	SPIROLOX PRB-131-M, or equal	See figure 1.
20	Packing, Class A, durometer 55±5	1	Rubber, ZZ-R-768	See figure 5.
21	Plate, identification	1	CRES or Al, Type I, MIL-DTL-15024	See figure 6.
22	Ring, retaining	1	M24231/5-076	See figure 1.
23	Wire, 15’	7	MIL-DTL-16878/36 BJE0	See figure 1 and figure 3.
24	Pin, positioning	1	M24231/5-054	See figure 7.
25	Epoxy resin, molded	As required	Cast epoxy resin compound 5/	See figure 3.

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TABLE I. Parts list – Continued.

NOTES:

1/ Item numbers are the same as encircled numbers shown on figures.

2/ See applicable figure or table listed in remarks column for part numbers and dimensions.

3/ Quantity is for one complete assembly of each size.

4/ Nitrile (Buna N); Parker Hannifin Corporation (part number N0756-75), O-ring Division, 2360 Palumbo Drive, Lexington, KY 40509, CAGE 02697, or equal.

5/ Arco Set 200, M.M. Reif and Co., Ivyland, PA, or PR610-TCF (Chockfast Orange), ITW Philadelphia Resins Corp., Montgomeryville, PA, or equal.

CHANGES FROM PREVIOUS ISSUE. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

Preparing activity:
Navy – SH
(Project 5935-2011-056)

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil.

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